Lecture 22

The critical vortex

A winding scalar phase cannot be made constant through its vortex core. At critical coupling, its static energy can be written as nonnegative squares plus a flux term. We keep the boundary contribution and specify the sector in which that flux has the required value.

Objectives. *We prove the critical energy bound with its signs and boundary assumptions. We derive the unit vortex's radial equations and regularity conditions. A smooth trial field shows why the same bound fails away from critical coupling.*

By the end of this lecture you will be able to:

- Derive the square completion for arbitrary smooth planar fields.
- Control the boundary current and identify the flux in the prescribed sector.
- Derive both radial equations with the one-form potential convention.

22.1 Planar energy and the imposed sector

Take the static model of Lecture 21, independent of the third spatial coordinate, with zero time and third potential components. Its energy is energy per unit length. On the oriented plane $\mathrm{d}x_1 \wedge \mathrm{d}x_2$, set $q = v = 1$ and write $A_{\mathrm{em}} = A_1\mathrm{d}x_1 + A_2\mathrm{d}x_2$, $D_j = \partial_j - \mathrm{i}A_j$, and $B = \partial_1 A_2 - \partial_2 A_1$.

Definition 22.1 (Planar Higgs energy)**.** For $\lambda > 0$, smooth real A_{em} and smooth complex Φ on $\mathbb{R}^2$, the energy is

$$E_\lambda = \int_{\mathbb{R}^2} e_\lambda \, \mathrm{d}^2x, \qquad e_\lambda = \frac{1}{2}\left[B^2 + |D_1\Phi|^2 + |D_2\Phi|^2\right] + \frac{\lambda}{8}(1 - |\Phi|^2)^2. \tag{22.1}$$

The *critical coupling* here is $\lambda = 1$.

Recall winding from *Topology and Geometry*, Part I, Lecture 3. A nonvanishing field on a circle has a phase lift along one traversal. Its phase increment is $2\pi n$ for an integer n, independent of the lift. The circle is traversed counterclockwise.

Definition 22.2 (Admissible winding sector)**.** For a fixed integer $n > 0$, the *admissible winding-n sector* consists of the smooth fields above with finite energy, $B \in L^1(\mathbb{R}^2)$, and $|\Phi(x)| \to 1$ uniformly as $|x| \to \infty$. On every sufficiently large circle, the phase loop has winding n.

Uniform convergence makes the field nonzero outside a disk, so these loops are defined and have the same winding by radial homotopy. The vacuum behaviour, integrable magnetic field and specified positive winding are hypotheses. Finite energy alone has not supplied them. For example, the constant vacuum has finite energy and winding zero.

22.2 The local identity and its boundary current

Define the real current one-form $K = K_j \mathrm{d}x_j$ by

$$K_j = \frac{\mathrm{i}}{4}\bigl(\Phi\,\overline{D_j\Phi} - \overline{\Phi} D_j\Phi\bigr) = \frac{1}{2}\,\mathrm{Im}(\overline{\Phi} D_j\Phi). \tag{22.2}$$

Write $\mathrm{curl}\,K = \partial_1 K_2 - \partial_2 K_1$. With $Z = \overline{D_1\Phi} D_2\Phi$, direct multiplication gives

$$|(D_1 + \mathrm{i}D_2)\Phi|^2 = |D_1\Phi|^2 + |D_2\Phi|^2 - 2\,\mathrm{Im}\,Z.$$

The commutator is $[D_1, D_2]\Phi = -\mathrm{i}B\Phi$. Using it when differentiating (22.2) gives

$$\begin{aligned} 2\,\mathrm{curl}\,K &= \mathrm{Im}\bigl[\overline{D_1\Phi} D_2\Phi - \overline{D_2\Phi} D_1\Phi + \overline{\Phi}[D_1, D_2]\Phi\bigr] \\ &= 2\,\mathrm{Im}\,Z - B|\Phi|^2. \end{aligned}$$

Expanding the magnetic square now proves the pointwise identity

$$e_1 = \frac{1}{2}|(D_1 + \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B - \frac{1 - |\Phi|^2}{2}\right]^2 + \frac{B}{2} + \mathrm{curl}\,K. \tag{22.3}$$

This calculation holds for arbitrary smooth fields, before imposing any equation. It recalls the completion of *Topology and Geometry*, Lecture 27; that Part II lecture is not assumed here.

We next control the two boundary integrals. Let $a(r) = \int_{|x|=r} |D\Phi|^2\,\mathrm{d}s$, where $|D\Phi|^2 = |D_1\Phi|^2 + |D_2\Phi|^2$. Finite energy gives $\int_1^\infty a(r)\,\mathrm{d}r < \infty$. On each interval $[e^j, e^{j+1}]$, logarithmic averaging provides a radius r_j with

$$r_j a(r_j) \le \int_{e^j}^{e^{j+1}} a(r)\,\mathrm{d}r \longrightarrow 0.$$

The modulus of Φ has a global bound M and is at least $1/2$ on these sufficiently large circles. Cauchy–Schwarz therefore gives

$$\left|\int_{|x|=r_j} K\right| \le \frac{M}{2}\sqrt{2\pi r_j a(r_j)} \longrightarrow 0.$$

Along a lifted phase ϑ on such a circle, $d\vartheta - A_{\mathrm{em}} = \mathrm{Im}(\overline{\Phi} D\Phi)/|\Phi|^2$. Hence

$$\left|2\pi n - \int_{|x|=r_j} A_{\mathrm{em}}\right| \le 2\sqrt{2\pi r_j a(r_j)} \longrightarrow 0.$$

Stokes' theorem from *Topology and Geometry*, Part I, Lecture 12, applies to these smooth one-forms on each compact disk, with counterclockwise boundary orientation. Since $B \in L^1$, it follows that

$$\int_{\mathbb{R}^2} B\,\mathrm{d}^2x = \lim_{j\to\infty} \int_{|x|=r_j} A_{\mathrm{em}} = 2\pi n. \tag{22.4}$$

The same subsequence makes the boundary current in the energy identity vanish.

Theorem 22.3 (Critical vortex bound)**.** *In the admissible winding-n sector at $q = \upsilon = \lambda = 1$, $E_1 \ge \pi n$. Equality holds exactly when*

$$(D_1 + \mathrm{i}D_2)\Phi = 0, \qquad B = \frac{1}{2}(1 - |\Phi|^2). \tag{22.5}$$

These are the Bogomolny equations.

Proof. Integrate (22.3) over the disks of radii r_j. Stokes gives the boundary integral of K. The proved limits of this boundary term and the magnetic flux give

$$E_1 = \pi n + \frac{1}{2}\int_{\mathbb{R}^2} \left\{ |(D_1 + \mathrm{i}D_2)\Phi|^2 + \left[B - \frac{1 - |\Phi|^2}{2} \right]^2 \right\} \mathrm{d}^2x.$$

The square integrals increase to their full-plane values. They are nonnegative, so the bound follows. Their sum is zero exactly when both continuous squared quantities vanish everywhere, which is (22.5). ∎

22.3 The unit radial vortex

Use polar coordinates (r, θ), with θ increasing counterclockwise. For real profiles f and α, the winding-one ansatz is

$$\Phi = f(r)e^{\mathrm{i}\theta}, \qquad A_{\mathrm{em}} = \alpha(r)\,\mathrm{d}\theta.$$

This is a one-form. Its component along the unit angular vector is α/r, not α. The Cartesian components are $A_1 = -\alpha \sin\theta/r$ and $A_2 = \alpha\cos\theta/r$. Differentiation gives

$$B = \frac{\alpha'}{r}, \qquad (D_1 + \mathrm{i}D_2)\Phi = e^{2\mathrm{i}\theta}\left[f' - \frac{1-\alpha}{r} f \right].$$

Thus both first-order equations reduce to

$$f' = \frac{1-\alpha}{r} f, \qquad \frac{\alpha'}{r} = \frac{1}{2}(1 - f^2). \tag{22.6}$$

For a smooth core, use the local forms $f(r) = rF(r^2)$ and $\alpha(r) = r^2 G(r^2)$ with smooth F and G. Indeed, then $\Phi = (x_1 + \mathrm{i}x_2)F(r^2)$ and $A_{\mathrm{em}} = G(r^2)(-x_2\mathrm{d}x_1 + x_1\mathrm{d}x_2)$ are smooth Cartesian fields. Writing $s_0 = f'(0)$, the equations give the regular expansions

$$f(r) = s_0 r - \frac{s_0}{8} r^3 + O(r^5), \qquad \alpha(r) = \frac{1}{4}r^2 - \frac{s_0^2}{8}r^4 + O(r^6).$$

The smooth core forms give $f(0) = \alpha(0) = 0$. We choose the positive vacuum branch $f(r) \to 1$ at infinity; the sector alone fixes its modulus. The flux identity (22.4) with $n = 1$ and radial Stokes give $\alpha(\infty) = 1$, since $\int B\,\mathrm{d}^2x = 2\pi[\alpha(\infty) - \alpha(0)] = 2\pi$.

Existence is a separate result. Taubes's Theorem 1 gives a smooth critical vortex for any prescribed finite multiset of zeros, whose multiplicities sum to n, unique up to gauge with the vacuum asymptotics. This is *cited*, not proved here: C. H. Taubes, *Communications in Mathematical Physics* **72** (1980), 277–292, Theorem 1, pp. 280–281, doi:10.1007/BF01197552. Its equations (2.6) have exactly the normalisation in (22.5).

Figure 22.1 illustrates the radial boundary problem. The numerically fitted initial slope is $s_0 \approx 0.603288$. The calculation illustrates the profiles and does not replace the cited existence theorem.

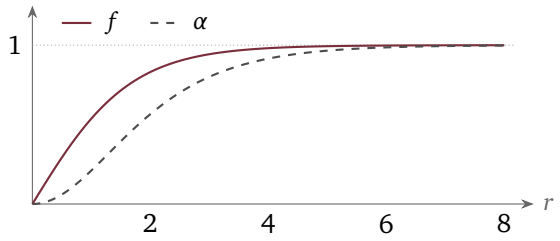


Figure 22.1: Numerical radial profiles: solid f and dashed α approach one. The calculation uses the regular origin series at $r = 10^{-4}$ and $f(18) = 1$; its normalized RMS collocation residual is below 10^{-8}.

22.4 The coupling is essential

For general positive λ, the completion has the additional term $(\lambda-1)(1-|\Phi|^2)^2/8$. A negative coefficient can destroy the critical bound. Consider the smooth winding-one trial

$$f(r) = \frac{r}{\sqrt{r^2+4}}, \qquad \alpha(r) = \frac{r^2}{r^2+4}.$$

Its Cartesian fields are $\Phi = (x_1 + \mathrm{i}x_2)/\sqrt{r^2+4}$ and $A_{\mathrm{em}} = (-x_2\mathrm{d}x_1 + x_1\mathrm{d}x_2)/(r^2+4)$, so they are smooth at the core. Its magnetic field is $8/(r^2+4)^2$ and its flux is 2π. The radial energy is

$$E_\lambda = 2\pi \int_0^\infty \left[\frac{1}{2}\left(\frac{\alpha'}{r}\right)^2 + \frac{1}{2}f'^2 + \frac{(1-\alpha)^2 f^2}{2r^2} + \frac{\lambda}{8}(1-f^2)^2\right] r\,\mathrm{d}r.$$

For this trial, $f' = (1-\alpha)f/r$. Substituting $s = r^2+4$ evaluates the integral exactly:

$$E_\lambda = \pi \int_4^\infty \left(\frac{32}{s^4} + \frac{16}{s^3} + \frac{2\lambda}{s^2}\right) \mathrm{d}s = \frac{2\pi}{3} + \frac{\pi\lambda}{2}.$$

At $\lambda = 1/4$ this is $19\pi/24 < \pi$. At the critical coupling it is $\pi + \pi/6$, since the trial does not satisfy the second Bogomolny equation away from the origin. A winding-one trial is not automatically an energy minimiser.

Summary. At $q = v = \lambda = 1$, the energy density is two nonnegative squares plus $B/2 + \mathrm{curl}\, K$. In the stated asymptotic sector, finite covariant-gradient energy supplies a sequence of large circles where the boundary current vanishes. The winding and integrable magnetic field give flux $2\pi n$. Thus $E \geq \pi n$, with equality exactly for the Bogomolny equations. The unit radial ansatz uses the one-form $\alpha\mathrm{d}\theta$ and has the stated regular core expansions. Existence is cited from Taubes. The bound need not hold away from critical coupling.

Self-check. Why is the flux contribution πn, rather than $2\pi n$, in this normalisation?

Answer. The local completion contains $B/2$. The total flux is $2\pi n$, so its energy contribution is $(1/2)(2\pi n) = \pi n$.

Self-check. For $A_{\mathrm{em}} = \alpha(r)\mathrm{d}\theta$, compute its unit angular component and the magnetic field. What core coefficient follows from the second radial equation?

Answer. They are α/r and $B = \alpha'/r$. Writing $\alpha(r) = a_2 r^2 + O(r^4)$ and $f(0) = 0$ gives $2a_2 = 1/2$, so $a_2 = 1/4$.

Self-check. Does finite energy alone select the positive winding sector and justify removing every large-circle boundary term?

Answer. No. The sector imposes the vacuum asymptotics, winding and magnetic integrability. The proof then produces a subsequence with vanishing boundary current. It does not assert vanishing along every radius from finite energy alone.

Self-check. Evaluate the smooth trial's energy at $\lambda = 1/4$. Does its unit flux force the critical lower bound at that coupling?

Answer. Its energy is $2\pi/3 + \pi/8 = 19\pi/24 < \pi$. The extra term in the off-critical completion is negative, so the critical bound does not apply even though the flux is 2π.